14 Notebook Computers in Brief

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Since the introduction of the first pocket computers just over two years ago and the first transportables about 1½ years ago, we have been avidly following the developments in the field. We have reviewed a cross section of these computers on the pages of *Creative Computing* (the dates of these reviews are noted in the chart at the end of the previous section).

This section is not intended to be an exhaustive, in-depth review of the computers. Rather, it contains a description of each machine including impressions from our review of it or from the comments of other users.

Texas Instruments CC-40

The Texas Instruments Compact Computer 40 (CC-40) bridges the gap between the pocket and notebook categories. It has one of the best versions of Basic ever offered by TI and provides exceptional accuracy, albeit rather slowly.

The CC-40 is powered by four AA alkaline batteries which will last for 200 hours of use, considerably longer than most other notebook machines. If you prefer, an AC adapter is also available.

Although the keyboard is arranged in the standard QWERTY layout, it is only two-thirds the size of a standard keyboard and sports calculator-style keys. Thus, touch typing is not possible, and even experienced typists will find a two-finger approach more reliable. For data entry, a numeric keypad is provided to the right of the main keyboard.

The CC-40 uses a single line, 31-character display capable of reproducing upper- and lowercase text and a variety of graphics symbols. The display scrolls horizontally on a maximum 80-character line. The LCD screen also displays several special status indicators above and below the main text line, so it is more versatile than it might appear.

As it comes out of the box, the CC-40 does not interface to anything directly—not even a cassette recorder. However, an eight-pin connector attaches to a hexbus peripheral module. This unit provides an interface to three peripherals, an RS-232 interface, a printer/plotter, and a wafertape drive. On the top of the CC-40 is a cartridge port that can accept ROM cartridge software or a memory expansion cartridge.

TI has announced a wide variety of software packages on both wafertape and ROM cartridges. The packages tend to be adaptations of programmable calculator software and will have greatest appeal for engineers and financial analysts.

The most likely market for the TI CC-40 is probably as a competitor to pocket computers and as an upgrade from programmable calculators. In this market with its low price tag (frequently discounted to under $200), it is a formidable competitor.

Casio FP-200

Casio is a very successful maker of calculators, watches, and electronic musical keyboards. However, their previous forays into computers have ended in failure, at least in the U.S. With the FP-200, they have taken a different approach and may be able to carve out their own niche in the market.

The FP-200 is primarily a spreadsheet machine and runs a built-in software package called CETL (Casio Easy Table Language). It is a *VisiCalc*-like language, and anyone familiar with another spreadsheet will be able to use it immedi-

Texas Instruments CC-40

Casio FP-200

WorkSlate

Canon X-07

ately. Also built-in is Casio Basic, a Microsoft-like implementation with rather leisurely performance.

The FP-200 is built around a CMOS version of the Z80 and has 32K of ROM and 8K of RAM, expandable to 32K. External mass storage requires either a cassette tape recorder (300 baud) or a 70K single-sided, single-density floppy disk drive. Output ports are also provided for a parallel printer and RS-232 serial device such as a modem.

The FP-200 has an 8-line x 20 character display. For graphics, 64 x 160 pixels can be individually addressed.

The keyboard is full-size, but uses calculator-style short-throw keys. It has 57 alphanumeric keys, four special keys, five function keys (two meanings each), and four cursor keys (arranged in a straight line). An optional, external numeric keypad is available.

The FP-200 would be a good choice if you are looking for a machine primarily for spreadsheet calculations and some Basic programming. The keyboard renders it unsuitable for word processing, even if there were software available which there is not. Several utility software packages (sort/merge, statistics, graphics) are promised and would further enhance the utility of the machine.

The suggested retail price is $499, but we have yet to see a machine in any retail outlets.

WorkSlate

Like the Casio FP-200, the WorkSlate from Convergent Technologies, is primarily a spreadsheet machine. Indeed, all the software packages on the WorkSlate are various adaptations of the basic spreadsheet program.

In a departure from all of the other notebook computers, the WorkSlate uses a CMOS version of the 6800 mpu. Included with the basic system is 16K of RAM memory. Currently, this is not expandable, but a company spokesman tells us that a 16K upgrade (32K total) will eventually be available. That is necessary, because the 16K gets filled up with just a 30 x 24 cell spreadsheet; in this state, it takes over two minutes just to perform a simple operation. Theoretically, a spreadsheet can expand to 128 rows and columns, but if you have 128 in either dimension, 5 is the maximum in the other.

The display on the WorkSlate consists of 16 lines by 42 characters. Some lines are devoted to status indicators, headings, and formulae; as a result, about 11 by 5 cells of a spreadsheet are visible at a time.

The keyboard follows the QWERTY layout, but it is about six percent smaller than a standard keyboard. This is not bothersome, but the circular keys with less than full travel might be. Clearly, it is not designed for text entry, and the majority of the numeric entries will be made from the numeric keypad to the right of the main keyboard. Indeed, there is no row of numeric keys over the alphabetic portion of the keyboard. A nice touch is the large diamond shaped cursor control key between the alpha and numeric keys.

To the right of the display is a microcassette recorder. This is a dual-function unit which can record either data or audio. The data transfer rate is 2400 baud, one of the fastest available on a small computer. For audio recording, WorkSlate has a built-in speaker, microphone, and jacks for an external earphone and microphone.

Also built in are a direct-connect, 300-baud modem and communications software. A matching printer, capable of 40- or 80-column compressed print widths is available for $250. Alternatively, an adapter is available for standard parallel or RS-232 serial printers or other devices.

We mentioned software packages other than the basic spreadsheet. These are Memo Pad, Phone List, and Calendar. Actually, the only difference between these and blank spreadsheets are some headings, graphics, and modifications of column width. In other words, a Memo Pad is simply a spreadsheet whose A column is 128 characters wide. Two self-instruction tutorial tapes are also available.

As the reviewer in our sister publication, *Computers & Electronics* concluded, "The features of WorkSlate are geared to the business person who wants spreadsheet capabilities in a portable package without all the fuss of learning about computers. For those who want such a business tool, WorkSlate has hit its design mark."

Canon X-07

The Canon X-07 is an exceptionally compact computer, weighing just over one pound. It has a four-line display, sensible cursor movement keys, and a novel optical communications capability.

The X-07 is primarily intended for computing in Basic, and no other software is currently built-in or offered. The default mode for calculations is double precision; as a result, the machine is quite accurate, but was the slowest of the 14 tested.

The X-07 uses a CMOS version of the Z80 mpu and has 20K of ROM and 8K of RAM (expandable to 24K). An external cassette recorder provides mass storage. The X-07 is powered by four AA akaline batteries with a rated life of "up to 2000 hours of use." Unbelievable!

Small memory cards about the size of two stacked credit cards are available with both ROM for applications software packages and RAM for removable user memory. Each card has a lithium watch battery which provides power for over a year.

The keyboard is laid out in QWERTY fashion, but it is 20% smaller than a standard keyboard and has calculator-

Hewlett Packard HP-75C

Toshiba T100

style keys. The spacebar is half the normal size, and there is only one SHIFT key, located below the Z. Thus, although it has a good feel, it is not suitable for rapid typing. However, for program editing, the cursor control keys are ideal, being oversize and laid out in a logical pattern.

The display has four lines of 20 characters each. Also, individual pixels can be addressed on the 32 x 120 screen. The X-07 also has a speaker capable of reproducing notes over four octaves.

Six peripherals are available for the X-07. A color plotter both prints and plots on 4½" wide paper in four colors. A compact thermal printer uses narrow (2¼") thermal paper. Three interface modules are available: one for RS-232 devices, one for a monitor or TV set, and another for interfacing to parallel devices such as Centronics-type printers, sensors, and synthesizers.

The last peripheral is an odd one; it is an infra-red optical coupler that permits two X-07 computers to communicate with each other. It makes a novel demonstration at trade shows, but we're not sure of what use it would be to the average user.

The Canon X-07 is a capable Basic-speaking computer in a compact package with a nice printer/plotter available. Support may be another issue. All of our testing of the X-07 was done in France and Japan where the Canon people were much more helpful than in the U.S. Buyers should hope that Canon will be more friendly to them than they are to the press.

Hewlett Packard HP-75C

The HP-75 is a very compact computer with unexpected speed and the highest accuracy of any computer we have ever tested—portable or desktop. It has a one-line display, but a monitor, as well as several other peripherals, are available.

At first glance, the keyboard appears to be much smaller than a standard one, but in fact is only 5% reduced. However, it uses calculator-style, short-travel keys and has a few keys in odd places; thus it is not suitable for rapid typing. A block of ten alpha keys can be specified as a numeric keypad for data entry.

Like earlier HP calculators, the HP-75 uses magnetic cards for program and data storage. Each card is 10" long and stores up to 1.4K. The cards must be pulled by hand through a slot on the lower right; it takes a bit of practice to get the hang of it, but the computer tells you if you have pulled it too fast or too slow.

The HP-75 uses a custom HP CMOS processor. It comes with 8K of RAM built in. Three slots on the front accommodate additional 8K RAM cartridges or ROM software cartridges. The machine can hold several programs which are accessed like a mini timesharing system. The HP-75 even has an Appointment Mode which can trigger one of nine different alarm sounds, a Basic program, or other action.

As expected, the HP-75 uses a dialect of HP Basic which is quite different from the DEC Basic, on which Microsoft Basic was modeled. Thus, string handling, functions, and PRINT USING (DISPLAY USING on the HP-75) are quite different from what much of the world regards as standard. Nevertheless, HP Basic is quite satisfactory and has several unexpected features such as powerful TRACE utility, recursive calls, and the ability to issue operating system commands from within a Basic program.

As might be expected, HP is converting programmable calculator application packages to the HP-75. Several engineering and financial packages have been released, and more are on the way. Even *VisiCalc* is available for the HP-75; with the single 32-character display line, it is less than satisfactory, but with the optional monitor hookup, it is fine. A word processing module has been announced, but we can't imagine doing serious text entry on the HP-75 keyboard.

The HP-75 uses an interface loop structure with each peripheral device daisychaincd in a continuous loop. Available peripherals include a digital cassette drive (fast and reliable), 24-column thermal printer, video interface (16 lines x 32 characters), plotter, 80-column dot matrix printer, and several laboratory device controllers.

The HP-75 is a fine computer system, particularly for a laboratory user or someone stepping up from a programmable calculator. It is well-engineered, has a good Basic (albeit a bit unusual), and a nice array of immediately available peripherals.

Toshiba T100

Unlike the other 13 machines in this roundup, the Toshiba T100 was not designed primarily as a notebook portable. Rather, it is the keyboard/system unit of an excellent desktop system to which attaches an eight-line LCD display and acoustic coupler to make it portable. It still needs AC power. Hence, it is primarily targeted at people who want to do all their computing on the same machine, but occasionally require computing capability away from the home office.

The T100 uses a Z80A running at 4 MHz; it was the fastest 8-bit machine we have tested to date. It has 32K of ROM and 64K of RAM built in. Two slots at the upper right accept ROM and RAM packs which are accessed from CP/M as disk drive E.

The keyboard is full-size with 89 full-stroke keys divided into a standard keyboard, numeric keypad, six special keys,

Teleram 3000

Radio Shack Model 100

and eight function keys (two user-programmable meanings each).

Three display connectors are provided, one each for a monochrome and RGB color monitor and one for the 8-line x 40-character LCD display. Individual pixels (64 x 240) can be addressed on the LCD display, a far cry from the spectacular 240 x 600 pixel resolution available on the color monitor.

For communications capability in either portable or stationary mode, the T100 uses the Lexicon LEX-12 modem, a 300-baud combination direct-connect/acoustic modem.

As mentioned, the T100 comes with CP/M 2.2 and Microsoft Basic in ROM. A disk, furnished with the portability package, scales the various menus to the LCD display and provides an assortment of utility packages. You can load any of these programs onto a RAM cartridge.

The T100 comes with an impressive array of bundled software including *Word Right, Magic Worksheet, MatheMagic, GraphMagic, Analyst, Q-Sort,* and *NAD.* Not all of these are suitable for use in the portable mode since several require disk operations.

The T100 has been on sale in Japan for over a year and is a proven performer. Toshiba offers a good array of peripherals including disk drives, monitors, printers, and a hardside carrying case for the cpu unit, display, and modem. The T100 should appeal to the user who wants, or can only afford, one computer; who needs an outstanding desktop machine; but who also needs limited away-from-home capability.

Teleram 3000

The Teleram 3000 was the first notebook portable introduced. It was targeted at large corporate users, and has found good acceptance in that market.

The 3000 is one of the largest notebook machines, weighing in at nine pounds. However, in this package, it has a full-size, full-stroke, completely standard keyboard; four-line by 80-character display; 128K of internal bubble memory (expandable to 256K); and CP/M operating system.

The 3000 has 64K of RAM; the one or two 128K bubble memory cartridges are accessed from CP/M like a floppy disk in drive A. The Teleram uses a CMOS version of the Z80; performance is similar to the four other machines using this configuration.

In addition to the standard alphanumeric keyboard, the 3000 has a numeric keypad to the right, eight programmable function keys with two meanings each, and several special keys.

Built into the 3000 are MBasic, several CP/M utilities, and a communications program called teleTalk. Microsoft Basic functions are as expected, although the graphics commands are not implemented; apparently the Teleram people couldn't imagine doing graphics on a 1.1" x 8.2" LCD screen—can't say as we blame them.

The teleTalk package is an especially rich communications package providing auto-dial, auto-answer, data capture, dump, and file transfer. Telephone numbers, passwords, commands, and log-on procedures may all be stored in command files. We were very impressed with the file transfer capabilities which allow all kinds of files, even CP/M .COM and text files, to be sent, received, and processed.

The Teleram 3000 has exceptional communications capabilities, a good Basic, and a standard operating system (CP/M) which opens the door to a large library of programs. Thus, the machine should have greatest appeal to the executive on the move, and the large company target audience of Teleram makes good sense.

Radio Shack Model 100

The story of the Radio Shack TRS-80 Model 100 is actually not the story of a TRS-80 at all. The machine was conceived by a small Japanese company, Kyoto Ceramics and first sold in Japan by NEC. Radio Shack designers worked with Kyocera to incorporate certain changes and additional features in the computer before introducing it in the American market. These changes were apparently "right," as few other computers have enjoyed such runaway sales success as the Model 100.

The Model 100 is truly notebook size (8.5" x 12" x 2.2") and weighs just under four pounds. It incorporates a full-size, full-stroke keyboard, with four special keys, eight function keys, and four cursor keys (in an unfriendly straight line).

The display is the largest on any notebook computer, 2" x 7.5", and displays eight lines of 40 characters each. The character size is large and legible. Graphics within a 64 x 240 pixel matrix are also possible. A built-in speaker plays notes over a five-octave range.

The Model 100 uses a CMOS version of the Z80 running at 2.5 MHz. Since default mode in Basic is double precision, the machine was very slow in running our benchmark; on the other hand, it scored high in the accuracy department. It has only 8K of RAM built in, but a 24K version is available. Both can be further expanded to 32K. An external cassette recorder provides mass storage.

The computer provides an impressive array of I/O ports. On the back are connectors for Centronics parallel printer, RS-232 serial device, cassette recorder, bar code reader, and modular telephone jack.

The Model 100 has a built-in direct-connect modem which can plug into any telephone jack. Coupled with the communications software package, it pro-

NEC PC-8201

Epson HX-20

vides many of the features of a so-called "smart" modem—auto-dial, log-on, download, and upload—although it does not have wake up and auto-answer.

The Model 100 has five programs built in. Microsoft Basic is missing a few commands and does not have on-screen editing (except by means of the text editor, a cumbersome process). The text editor is an adequate package. It is always in insert mode, and has cut, paste, search, and other rudimentary features. It does not have an output formatter, but several are available from third party vendors.

The communications package was mentioned above. The last two packages, schedule organizer and name/address organizer, are simply special versions of the text editor with certain commands locked out. We have not found them particularly useful.

Many software packages have been introduced already by third party vendors, and much more is on the way. The availability of software coupled with the integrated packages built into the machine make the Model 100 an attractive choice for a wide variety of users. Poor Basic program editing and lack of an output formatter are small drawbacks against the many enticing capabilities of the computer coupled with an attractive price.

NEC PC-8201

We have frequently called the NEC PC-8201 the twin of the Radio Shack Model 100. Strictly speaking, this is not true. The 8201 was born six months or so earlier in Japan and is a somewhat different version of the Kyoto Ceramics original.

The 8201 is slightly larger than the Model 100 as a result of providing a slot in the left side for an expansion memory cartridge. The 8201 can have from 16K to 64K of RAM in the basic machine. The 32K plug-in memory cartridges function as a switchable bank of main memory, rather than a disk drive as on some other machines.

The 8201 has a full-size, full-stroke keyboard with two special keys, five function keys (two programmable meanings for each), and cursor control keys laid out in a logical diamond pattern. Graphics characters are not built into the 8201 as they are on the Model 100; instead any desired graphics characters can be entered by the user with a short included utility program.

The display is 2" x 7.5" and displays eight lines of 40 characters each. The version of Microsoft Basic on the NEC has the LOCATE command not found on the Model 100, so characters as well as individual pixels (64 x 240) can be addressed. The Basic on the NEC also has on-screen editing and RENUM, both lacking in the Model 100 implementation.

In addition to Basic, the 8201 has a text editor (but no output formatter) and telecommunications software package (but no built-in modem). The 8201 also comes with a cassette tape of 16 utility and demonstration programs—a desirable extra.

The 8201 has output connectors for Centronics parallel printer, RS-232 serial device, bar code reader, modem, and cassette recorder. The literature promises a floppy disk interface, but we have been unable to get any information on it.

For an attractive price, the NEC PC-8201 offers a great deal of capability—Basic computations, word processing, and communications, with RAM cartridges for external storage. The machine should have appeal for a wide cross section of people needing a true portable computer.

Epson HX-20

The Epson HX-20 was the first true notebook size computer introduced. Unfortunately, lack of availability prevented it from being a runaway success when it was introduced in late 1982. Now that it is widely available, it no longer has the market to itself and will have to carve out a smaller niche among a tough field of competitors.

Several reviewers have looked at the small screen size of the HX-20 and concluded that it is not competitive with the later entries sporting screens four to eight times larger. We think that is unfair, as the HX-20 still offers a wide array of features, some of them unique in a machine this size.

The Epson is slightly thinner than the Model 100, but weighs the same 3.8 pounds. It has a full-stroke, standard size keyboard with an excellent feel. Along with the 54-key QWERTY keyboard, it has seven special keys and five function keys, each with a dual meaning. The main disadvantage is that there are only two cursor control keys; the other two directions are gotten by pressing shift with one of the two keys. Although Basic has on-screen editing, using just two keys is a pain.

The LCD screen displays four lines of 20 characters each. The display is actually a window onto a much larger virtual screen; the size can be specified by the user. Hence, it is possible to scroll in both directions. Pixel and character addressing are possible within the 32 x 120 pixel dimensions of the screen. A small speaker can produce tones over a four-octave range.

The HX-20 uses a CMOS version of the Z80 mpu. It has 32K of ROM and 16K of RAM, expandable to 32K with an external module. Mass storage is provided in the form of a built-in microcassette recorder. We found this to be fast and reliable. An external cassette can also be used.

The HX-20 provides I/O connectors for RS-232 serial devices, bar code reader, cassette recorder, and a 38,400-


RoadRunner


Sharp PC-5000

baud serial link to other devices via an interface module which has yet to be released.

On the top left of the case is a built-in printer. It uses plain paper rolls 2¼" wide and prints in black or purple. It is this printer that makes the small screen size tolerable as programs or text can be printed out in rough form for correction and then printed later on a full-size printer or transmitted to another machine.

The NiCad rechargeable battery on the HX-20 provides 50 hours of use, considerably more than any of the other notebook portables.

Built into ROM is a rudimentary monitor, Microsoft Basic, and *Ski-Writer,* a word processing package. Basic is a complete implementation with no obvious omissions. Up to four Basic programs can be stored simultaneously in the machine. Many more, of course, can be stored on tape.

SkiWriter is an adequate, if not extensive, word processing package. It can operate in either insert or overstrike mode and has block copy and delete. It will search for a string, but will not search and replace automatically. Print formatting is barely adequate, as it requires that you put page breaks into the text rather than producing them automatically.

Epson has announced a wide array of plug-in ROM software packages, but we have not seen them at the retail level yet. The communications and spreadsheet packages should enhance the appeal of the machine considerably.

The HX-20 with built-in printer and optional microcassette offers a great deal of computing power at a moderate price. The long battery life between charges is a nice plus. The machine should appeal to people needing a full-feature Basic and occasional word processing. Additional software packages should help it carve out a niche with specific types of users.

RoadRunner

The MicroOffice RoadRunner is one of the latest entries in the notebook computer sweepstakes. Currently, it is being marketed primarily to OEMs and large-volume end users, but it may be available by mail order and in a limited number of retail computer stores.

The RoadRunner is equal in size to a large binder and weighs in at five pounds. When it is opened, it comes to life with a small beep and initial dialog on the display.

The LCD screen measures 1.3" x 9.3" and displays eight lines of 80 characters, about the size of a dot matrix printer. Graphics can be displayed on the 64 x 480 pixel screen. The display tilts, but unfortunately does not have a contrast adjustment.

The keyboard is a full-size, full-stroke unit with sculpted concave keys. Although close to a standard layout, it has several keys in unexpected locations. Most users will probably adjust in a week or two. The cursor keys are laid out in a logical pattern, an oversight on too many notebook portables. Six special keys and eight dual-meaning function keys are found in the top row above the standard QWERTY keyboard.

The RoadRunner uses a CMOS version of the Z80 mpu and has 16K of ROM and 48K of RAM. Four memory cartridge slots are found over the keyboard for extra RAM memory and ROM software cartridges. These are addressed from the CP/M-compatible operating system as devices A through D.

I/O connectors are provided for RS-232 serial devices, a modem module, and the main bus of the computer. The modem is a 300-baud direct-connect unit with auto-dial, auto-answer, and a wake-up mode of operation. In addition, the RoadRunner has a built-in terminal mode which emulates a DEC VT100 terminal.

Also built in is a schedule organizer, name/address organizer, and, of course, the CP/M operating system. Available on cartridge is a full-feature word processing package with features such as character, word, and line delete, and global search-and-replace.

Cartridges are also available with Microsoft Basic (with everything but on-screen editing) and Sorcim *SuperCalc.* More packages are promised in the future.

With its excellent communications, word processing, Basic, and spreadsheet software on a machine with full keyboard, large display, and plug-in memory cartridges, the RoadRunner is sure to find a market with executives, sales people, and writers who need a portable office on the road.

Sharp PC-5000

The Sharp PC-5000 is one of the largest of the notebook computers, but it is packed with features and capability. It has a large screen (eight lines by 80 characters), full keyboard, 16-bit processor, 128K memory, and much more.

The PC-5000 uses a 16-bit 8088 mpu, the same as in the IBM PC. MS-DOS

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Xerox 1810

and Microsoft GW (Gee Whiz) Basic reside in 64K of ROM, and 128K of RAM is available for user memory, expandable to 256K. Mass storage is in the form of a 128K bubble memory cartridge or, if you prefer, an external cassette recorder. In a non-portable mode, the PC-5000 also supports a double-density, double-sided floppy disk drive.

With the 16-bit processor, the PC-5000 is fast—close to the fastest computer we have ever tested, portable or not. A second control mpu controls the I/O functions, further contributing to the throughput.

The LCD screen measures 1.3" x 9.3" and displays eight lines of 80 characters or graphics in an 80 x 640 pixel field. Characters are about the same size as dot matrix printer output—small, but readable.

The keyboard has 57 full-stroke keys, three special keys, eight dual-meaning function keys, and four cursor control keys (laid out in a straight line, unfortunately). The keys are concave sculpted, and have a good feel, except for a slight "give" in the center of the keyboard.

Connectors are provided for a cassette recorder, external bus, RS-232 serial device, and modem. The modem is an option with the PC-5000 and fits into the lid of the case. It is a 300-baud, direct-connect unit with an auto-dial, redial, and conference phone capability. It is supported by the *SuperComm* software package by Sorcim.

Another optional extra which fits in the basic unit is an 80-column thermal printer which can print on plain paper as well as on thermal paper. It is a 30 cps unit that produces excellent type (and dot graphics) in a variety of formats.

Software is available on bubble memory cartridges or on disk. In addition to the built-in MS-DOS and Microsoft GW Basic, external software includes *SuperWriter*, an excellent menu-driven word processing package by Sorcim; *SuperCalc-2*, a spreadsheet package; and *SuperComm*. Many other software houses are said to be working to fit their packages on the PC-5000.

The PC-5000 is an outstanding, full-featured computer with a variety of excellent software backed up by some innovative serving arrangements. As such, it should find enthusiastic acceptance by a wide cross section of business people and other people on the move who need full-function computing power.

Xerox 1810

The Xerox 1810 is a notebook portable designed by Sunrise Systems. It is sold (by them) only to OEMs. Currently, the only OEM actively marketing the system is Xerox; from them it is known as the Xerox 1800. It is likely to be available from other vendors in slightly different configurations in the near future.

Like the Toshiba T100, the 1810 is the keyboard, cpu, and system unit of a desktop computer, although the LCD display is built in. The machine can be customized to the particular needs of an individual vendor, thus some vendors may offer it as a desktop unit with portable cpu, and others may offer it just as a notebook portable. Xerox offers both the 1810 keyboard unit plus an optional 1850 "Flat Pack" processor.

The processor is a CMOS version of the Z80 with 32K of ROM and 16K of RAM, expandable to 64K. In the desktop configuration, the unit contains a dual 16-bit 8088 and an 8-bit Z80 plus additional memory. Mass storage in the portable mode of operation is provided by a built-in microcassette recorder to the right of the keyboard. This stores 512K of digital information and also doubles as an audio recorder.

The keyboard has 63 full-stroke keys in a more or less standard layout. The

Gavilan

cursor control keys are arranged logically, although the up cursor key is where you might expect to find the question mark on the bottom row. A row of special and programmable function keys is over the regular keyboard.

The LCD display is available in two versions: six lines by 40 characters or three lines by 80 characters. For some inexplicable reason, Xerox chose the latter configuration for the 1810. Individual pixels are addressable, although hardly useful in the 3 x 80 layout (24 x 480 pixels). The unit also provides a signal for a monitor.

A direct-connect 300-baud modem, supported by a dumb terminal software package, is built in. An interesting telephone program lets the computer be used as an answering machine, and provides auto-dial, redial last number, hold, and speakerphone capabilities.

I/O connectors are provided for RS-232 serial devices, Centronics parallel printer, telephone, and RGB or monochrome monitor.

A vast array of software is available and under development. Built in is CP/M, a calendar/scheduler, and four-function calculator. Software on plug-in ROM cartridges includes Microsoft Basic, typewriter/note taker (a word processing package), a terminal program, and the telephone program mentioned above.

The Xerox 1810 is a state-of-the-art machine with a wide array of software and easy access to much more as a result of the CP/M operating system and communications capability. We will be anxious to see how Xerox (and other vendors) target it.

Gavilan

When the Gavilan was first shown at NCC in June 1983, many people wondered out loud whether it could really be made. It was just too much state-of-the-art stuff in one package for people to swallow. Now, that it is nearing readiness for the market and now that competitors are leaping in from all directions, the Gavilan looks more real than it did six months ago. But it still has an air of unreality.

The Gavilan is remarkably compact (11.4" square x 2.7" high) and weighs just nine pounds. Its optional, snap-on printer adds four inches and four pounds to the package. It has a 16-bit mpu, 64K of RAM, built-in 3" floppy disk, eight-line display, full keyboard, and a unique touch panel in which your finger becomes sort of an electronic mouse.

The Gavilan uses a 16-bit 8088 mpu, 48K of ROM, and 64K of RAM, expandable with up to four 32K plug-in capsules of blank memory or applications software packages. Also built in is a 3", 320K Hitachi floppy disk drive. Traffic cop chips turn off the power to the disk drive or mpu whenever it is not needed; thus the NiCad batteries provide eight hours of operation. Eighty percent of the charge can be restored in one hour on the charger.

The keyboard is a standard-size, full-stroke unit with 13 special keys to the right. Several keys are in "unusual" locations, especially the enter key next to the spacebar, but it shouldn't take too long to get used to it.

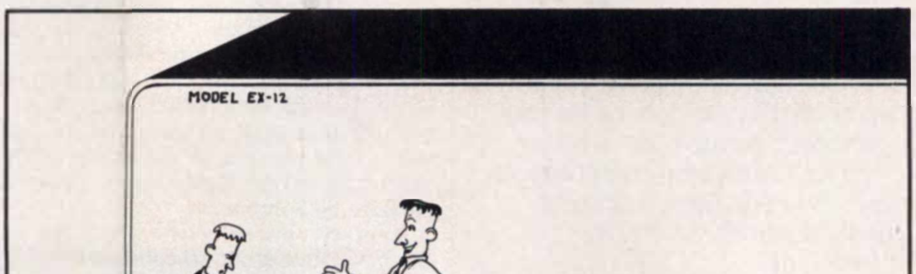
The most unusual feature of the Gavilan is the touch pad below the display. This lets you manipulate objects on the screen by pointing at them. A quick movement of your finger moves the cursor a long way while a slow movement gives you fine control. Like Apple's Lisa system, pictorial representations of objects such as file drawers, file folders, documents, and a trash basket are shown on the screen.

Although the screen is capable of displaying eight lines of 80 characters, in most cases, part of the screen will be devoted to menus in "windows" appropriate to the software package currently in use. This loss of a portion of the screen makes one wish mightily for a screen three times as large. At home base your prayers are answered, since the Gavilan provides standard video output to a monochrome monitor.

The Gavilan has a built-in 300-baud direct-connect modem supported by a comprehensive software package. It also has interfaces for the optional printer, a second disk drive, RS-232 serial device, and video output.

Besides MS-DOS, MBasic, and the communications software, Gavilan makes available an Office Pack of four applications, Sorcim *SuperCalc* and *SuperWriter*, and *PFS File* and *Report*.

All in all, the Gavilan offers as much or more than most desktop systems. It is a full-function computer with few trade-offs. All this comes at a price, but for the traveling professional this is a machine that is easy to learn and exceptionally user-friendly, and one that will spark envy in the eyes of all who see it. ■



"... and once you're ready to use this computer with software all you'll need is the expansion module."